NEXUS TERMINAL / PCF3 ARCHITECTURE BRIEF

System Architecture, Data Provenance & Decision Layer Integration Plan (Grok AI Collaboration)

Executive Summary: This technical specification outlines the repository layout, automated ingestion pipelines, data-mode labeling, and gate architectures of the Nexus Terminal / PCF3 system. It details the optimal path to incorporate a constrained, non-discretionary **Decision Layer** on top of the existing 9-Desk Research Packet without altering live telemetry collection or the 4H automated delivery pipeline.

1. High-Level Repository Structure & Core Entry Points

```
pcf3-v9-nexus/
  .github/workflows/collector.yml       # GitHub Actions 4H & Watchdog runner (cron: '5,35 * * * *')
  pcf3_collector_default.py             # Core Python collector (generates PCF3_LIVE_PACKET_DEFAULT.txt)
  PCF3_LIVE_PACKET_DEFAULT.txt          # The canonical 70-card baseline market packet
  scripts/
    macro_collector.py                  # Macro feed collector (SPX, RSP, US10Y, MSTR -> macro_threat_feed.json)
    send_telegram_master_packet.py      # Unified 9-Desk dossier assembly & Telegram bot dispatcher
  data/
    macro_threat_feed.json              # Macro breadths, 10Y yield, MSTR canary, composite threat score
    volatility_confluence_feed.json     # Desk 8 volatility radar metrics & regime headwinds
    telegram_last_state.json            # State tracking (last4hInterval, prices, locks, alert triggers)
    oi_history.json                     # Perpetual open interest historical trajectory
```

- **Data Ingestion:** pcf3_collector_default.py & scripts/macro_collector.py
- **Packet Assembly & Dispatch:** scripts/send_telegram_master_packet.py via build_unified_master_dossier()

2. On-Demand (/now) & Automated 4H Telemetry Flow

• **Trigger Mechanisms:**
- **On-Demand (/now):** Triggers collector.yml via GitHub Actions workflow_dispatch with FORCE_TELEGRAM_SEND=true.
- **Automated 4H Close:** Scheduled at 5,35 * * * * (5 and 35 minutes past every hour). Running at :05 past the hour executes immediately after the 4H candle closes with zero queue congestion. send_telegram_master_packet.py compares current_4h_interval against last4hInterval in data/telegram_last_state.json.

• **Unified Assembly (send_telegram_master_packet.py):**
1. Live spot price & order book depth polled from Binance Vision API (Coinbase fallback).
2. Daily & 4H klines compute 200 SMA slope, 21 EMA distance, 14-day RSI, 20-day ATR, and 30-day Volume POC.
3. Ingests macro_threat_feed.json, volatility_confluence_feed.json, and PCF3_LIVE_PACKET_DEFAULT.txt.
4. Generates NEXUS_9DESK_MASTER_PACKET_*.txt (~50 KB) containing the system prompt, Desks 1–9, and 58-card Appendix A.
5. Transmits document via Telegram Bot API with an executive HTML caption strictly under 1,015 characters.

3. Data-Mode Labeling Architecture

Located in pcf3_collector_default.py (Lines 72–128):
• **Status Taxonomy:** LIVE | CACHED | DERIVED | PROXY | STALE | MISSING | MOCK
• **Method Taxonomy:** OBSERVED | CALCULATED | PROXY | CACHED | ESTIMATED | UNAVAILABLE
• **Confidence Metric:** HIGH | MEDIUM | LOW
• **Missing-Data Governance Gates (apply_missing_data_gates):**
- Any MOCK active → *ANALYSIS ONLY — DO NOT TRADE*.
- ≥ 2 critical metrics missing → *STAND ASIDE*.
- 1 critical metric missing → Sizing capped at *0.5x*.

• **Unified Dossier Provenance Ledger:** L1: Live Stream, L2: Derived Indicators, L3: On-Chain & Macro, L4: Strategy & Models.

4. Threat Matrix & Desk 7 CRIMSON Gate Logic

Located in scripts/macro_collector.py (Scoring) and scripts/send_telegram_master_packet.py (Gate Enforcement):

```
# 1. Macro Threat Scoring Breakdown (0 to 100 points):
#    • Module A: SPX / RSP Breadth Divergence (0 to 35 pts)
#    • Module B: US 10Y Yield Elevation (0 to 30 pts) [Threshold: >= 5.0%]
#    • Module C: MSTR Bearish Divergence vs BTC (0 to 20 pts)
#    • Module D: Pre-Election / Seasonal Window (0 to 15 pts)

# 2. Gate Condition:
is_crimson = (threat_score >= 66 or threat_level == 'CRIMSON')

# 3. Capital Preservation Precedence Lock:
if is_crimson:
    verdict_lock = 'FLAT_100_USDT | MAX_EXPOSURE: 0.00x | GATE_DRIVER: DESK_7_CRIMSON'
    effective_dampener = 0.00
    # Lower desk triggers (dip-buying rungs, quant confluence, SMC sweeps) are HARD-LOCKED
```

5. Recommended Implementation for the Decision Layer

Recommendation: Dual-Layer Output (Header Checkpoint + Telegram Mobile Summary)
This approach guarantees 100% data integrity for the Research Packet while giving human traders and LLMs an instant, non-discretionary checklist.

A. Top-of-Packet Decision Checkpoint (In the generated .txt):
Prepend a standardized checklist block directly above Section 1 of the master dossier:

```
================================================================================
NEXUS DECISION LAYER — CAPITAL PRESERVATION CHECKLIST
================================================================================
[1] DESK 7 CAPITAL PRESERVATION GATE : [MET / ACTIVE] (Threat 85/100 CRIMSON)
    -> OVERRIDE TRIGGERED: Mandates 100% USDT / 0.00x Notional Exposure.
[2] THREE CRITICAL ANCHORS:
    • BTC Live Spot Price           : $81,529.92 [CONFIRMED LIVE]
    • US 10Y Treasury Yield         : 5.098%     [ELEVATED >5.0%]
    • STH Realized Cost Basis Floor : $74,468.34 [BUFFER +9.48%]
[3] SYSTEM DECISION VERDICT         : [STAND ASIDE]
    • Classification               : CASH PRESERVATION / RISK OFF
    • Active Exposure Cap          : 0.00x
    • Directive                    : Capital preservation overrides all setups. Zero buy/sell signals.
================================================================================
```

B. Telegram Mobile Summary (In the Telegram caption):
Update the LLM AI ESCALATION ADVISOR block inside send_telegram_master_packet.py to display the exact 3-point checklist directly on mobile. Keeps captions under the 1,024-character Telegram limit.

Key Advantages of this Strategy:
1. **Zero Regression Risk:** No GitHub Actions workflow or collector changes needed.
2. **Single Source of Truth:** Existing 9-Desk Research Packet is preserved byte-for-byte in the same file.
3. **Zero Execution Ambiguity:** Fully deterministic checklist; strictly avoids speculative trade recommendations.